

DATA STRUCTURES AND ALGORITHMS (DSA) - ASSIGNMENT 01 ANSWERS

Prepared for GitHub Submission

1. Complexity Ranking: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^5)$
- 2a. $O(n^2)$
- 2b. $O(n^3)$
- 2c. $O(n^3)$
- 2d. $O(n^2)$
- 2e. $O(n^3)$
- 2f. $O(n)$
3. Algorithm runs in $O(n)$ using a loop from 1 to n.
- 4a. $O(\text{number})$
- 4b. $O(\text{cycles})$
- 4c. $O(\text{iter}^2)$
- 4d. $O(\text{iter}^2)$
- 5(i). $\leq$
- 5(ii). $\geq$
6. $4n + 12$ is $O(n)$, choose $c=5$ and $k=12$.
7. $5n^2 - 16$ is $O(n^2)$, choose $c=5$ and $k=1$.